Relativistic Energy

PHYS 2203 — Introductory Physics III

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Overview

Last lecture showed that momentum conservation in every frame needs a partner: the quantity $\sum_i \gamma m_i$ must be conserved too. This lecture identifies that partner as energy. Computing the work needed to accelerate a particle from rest gives the relativistic kinetic energy $K = (\gamma - 1)mc^2$, and with it the total energy $E = \gamma mc^2$, which includes a rest energy mc^2 even for a particle that is not moving. Energy and momentum then combine into a four-vector whose invariant is the mass, through $E^2 = p^2c^2 + m^2c^4$. We close with an important consequence: a system's mass need not equal the sum of its parts' masses.

1 Work and Energy in Relativity

Last time we found that if a system of particles has total relativistic momentum $P_{\text{total}} = \sum_i \gamma_{u_i} m_i u_i$ in frame O, then in a frame O' moving at speed v relative to O its total momentum is

$$\boxed{P'_{\text{total}} = \gamma_v P_{\text{total}} - v\,\gamma_v \sum_i \gamma_{u_i} m_i.} \tag{1}$$

Suppose a collision leaves P_{total} unchanged in O. Then P'_{total} is unchanged in O' only if $\sum_i \gamma_{u_i} m_i$ is unchanged as well. So momentum conservation in every frame requires a second conserved quantity, $\sum_i \gamma_{u_i} m_i$. We will derive what this term is by calculating how much work is needed to accelerate a particle from rest, which should give us an expression for the kinetic energy.

The work done by a force F to move an object a distance $\mathrm{d}x$ is

$$\mathrm{d}W = F\,\mathrm{d}x = \frac{\mathrm{d}p}{\mathrm{d}t}\,\mathrm{d}x. \tag{2}$$

The rate of work done is

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}p}{\mathrm{d}t}\,\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}p}{\mathrm{d}t}\,u, \tag{3}$$

and changing the differentiation variable from t to u,

$$\frac{\mathrm{d}W}{\mathrm{d}u}\,\frac{\mathrm{d}u}{\mathrm{d}t} = \frac{\mathrm{d}p}{\mathrm{d}u}\,\frac{\mathrm{d}u}{\mathrm{d}t}\,u \qquad \Longrightarrow \qquad \frac{\mathrm{d}W}{\mathrm{d}u} = \frac{\mathrm{d}p}{\mathrm{d}u}\,u. \tag{4}$$

Integrating from rest to u_f

Now accelerate the object from rest up to a final speed u_f, and add up all the work done along the way:

$$W = \int_0^{u_f} \frac{\mathrm{d}W}{\mathrm{d}u}\,\mathrm{d}u = \int_0^{u_f} u\,\frac{\mathrm{d}p}{\mathrm{d}u}\,\mathrm{d}u = pu\Big|_0^{u_f} - \int_0^{u_f} p\,\mathrm{d}u, \tag{5}$$

The boundary term. This one needs no work at all, since $p = \gamma m u$:

$$pu \Big|_0^{u_f} = \gamma\, mu^2 \Big|_0^{u_f} = \gamma_f\, mu_f^2, \tag{6}$$

because both u and p vanish at the lower limit.

The remaining integral. Writing p out in full,

$$\int_0^{u_f} p\,\mathrm{d}u = \int_0^{u_f} \frac{mu}{\sqrt{1 - u^2/c^2}}\,\mathrm{d}u. \tag{7}$$

The integrand is awkward only because of the square root, and the substitution

$$s \equiv 1 - \frac{u^2}{c^2} \tag{8}$$

removes it. Differentiating the substitution,

$$\mathrm{d}s = -\frac{2u}{c^2}\,\mathrm{d}u \qquad \Longrightarrow \qquad u\,\mathrm{d}u = -\frac{c^2}{2}\,\mathrm{d}s. \tag{9}$$

Substituting,

$$\begin{aligned}
\int \frac{m\,u\,\mathrm{d}u}{\sqrt{1 - u^2/c^2}} &= \int \frac{m}{\sqrt{s}} \left(-\frac{c^2}{2}\,\mathrm{d}s\right) \\
&= -\frac{mc^2}{2} \int s^{-1/2}\,\mathrm{d}s \\
&= -\frac{mc^2}{2} \left(2\, s^{1/2}\right) \\
&= -mc^2 \sqrt{1 - \frac{u^2}{c^2}} \\
&= -\frac{mc^2}{\gamma},
\end{aligned} \tag{10}$$

where the last line uses $\sqrt{1 - u^2/c^2} = 1/\gamma$. So the antiderivative of $-p$ is $+mc^2/\gamma$, and evaluating between the limits (where $\gamma = 1$ at $u = 0$),

$$-\int_0^{u_f} p\,\mathrm{d}u = \frac{mc^2}{\gamma} \Big|_0^{u_f} = \frac{mc^2}{\gamma_f} - mc^2 = mc^2 \left(\frac{1}{\gamma_f} - 1\right). \tag{11}$$

Since the object started from rest, all of this work went into its motion. By the work–energy theorem the total work done *is* the kinetic energy the object ends up with:

$$W = K. \tag{12}$$

Relativistic kinetic energy

Plugging in,

$$\begin{aligned} K &= mc^2\left(\frac{1}{\gamma_f} - 1\right) + \gamma_f\, m u_f^2 \\ &= \gamma_f\, mc^2\left[\frac{1}{\gamma_f^2} - \frac{1}{\gamma_f} + \frac{u_f^2}{c^2}\right] \\ &= \gamma_f\, mc^2\left[\left(1 - \frac{u_f^2}{c^2}\right) - \frac{1}{\gamma_f} + \frac{u_f^2}{c^2}\right] = \gamma_f\, mc^2\left[1 - \frac{1}{\gamma_f}\right], \end{aligned} \tag{13}$$

where $\gamma_f = 1/\sqrt{1 - u_f^2/c^2}$ and we used $1/\gamma_f^2 = 1 - u_f^2/c^2$. Dropping the subscript f,

$$\boxed{K = \gamma\, mc^2 - mc^2 = (\gamma - 1)\, mc^2} \qquad \text{(relativistic kinetic energy)}. \tag{14}$$

If $u \ll c$, then $\gamma \approx 1 + \dfrac{u^2}{2c^2}$ (the first two terms of the binomial expansion), so

$$K \approx mc^2\left(1 + \frac{u^2}{2c^2}\right) - mc^2 = \tfrac{1}{2} m u^2. \tag{15}$$

The classical relation $K = \frac{1}{2} m u^2$ is an excellent approximation when $u \ll c$.

As $u \to c$, the kinetic energy of a massive particle becomes infinite. Hence, it would require infinite work to accelerate a massive particle to the speed of light, and we see yet again why c is the speed limit of the universe.

2 Total Relativistic Energy and Mass–Energy

Let us define a **total relativistic energy** E. Starting from

$$K = \underbrace{\gamma\, mc^2}_{\text{total relativistic energy}} - \underbrace{mc^2}_{\text{rest energy}}, \tag{16}$$

we identify

$$\boxed{E = K + mc^2 = \gamma\, mc^2.} \tag{17}$$

Setting $u = 0$ gives $E = mc^2$. An object sitting still has energy, set by its mass alone. This is the **rest energy**, of a particle, and give us Einstein's most famous expression.

Now go back to Eq. (1), the transformation of the total momentum:

$$P'_{\text{total}} = \gamma_v P_{\text{total}} - v\, \gamma_v \sum_i \gamma_{u_i} m_i. \tag{18}$$

Since $E_i = \gamma_{u_i} m_i c^2$, the sum is

$$\sum_i \gamma_{u_i} m_i = \frac{1}{c^2} \sum_i E_i = \frac{E_{\text{total}}}{c^2}, \tag{19}$$

where E_{total} is the total energy of the system in frame O. Substituting,

$$\boxed{P'_{\text{total}} = \gamma_v \left(P_{\text{total}} - \frac{v}{c^2} E_{\text{total}} \right).} \tag{20}$$

If a collision leaves both P_{total} and E_{total} unchanged in O, then P'_{total} is unchanged in O'. So momentum is conserved in every frame as long as energy is also conserved. Requiring momentum conservation to hold in every frame forces energy conservation.

For a system of N particles,

$$\sum_{i=1}^{N} E_i \quad \text{is unchanged by collisions and reactions between particles} \tag{21}$$

(**conservation of total relativistic energy**).

3 The Energy–Momentum Relation

We now derive the relationship between E and p. Starting from $E^2 = \dfrac{m^2c^4}{1 - u^2/c^2}$ and adding and subtracting $m^2u^2c^2$ in the numerator,

$$\begin{aligned} E^2 &= \frac{m^2c^4}{1 - u^2/c^2} = \frac{m^2u^2c^2 + m^2c^4 - m^2u^2c^2}{1 - u^2/c^2} \\ &= \frac{m^2u^2c^2}{1 - u^2/c^2} + \frac{m^2c^2(c^2 - u^2)}{1 - u^2/c^2} = \frac{m^2u^2c^2}{1 - u^2/c^2} + \frac{m^2c^4\,(1 - u^2/c^2)}{(1 - u^2/c^2)}. \end{aligned} \tag{22}$$

The second term reduces to m^2c^4 and the first is p^2c^2, giving

$$\boxed{E^2 = p^2c^2 + m^2c^4} \qquad \text{where} \qquad p = \gamma\, mu. \tag{23}$$

For particles with small m and/or high u, we can approximate $E \approx pc$ (the "extreme relativistic approximation").

4 The Energy–Momentum Four-Vector

In Lecture 14 we introduced the idea of a **four-vector**: any four quantities that transform between inertial frames by the Lorentz transformation. The example there was the position-time four-vector (x, y, z, ct). We now see how momentum and energy combine to form a different type of four-vector.

Notation for this section. As in Lecture 17, we distinguish two Lorentz factors: v is the relative speed of the two *frames*, with boost factor γ_v, while u is the particle's own speed within a frame, with factor γ_u. Elsewhere in this lecture only the particle's speed is in play, and the bare γ means γ_u.

How energy and momentum transform

Let a particle have momentum components p_x, p_y, p_z and total energy E as measured in frame O. Starting from the definitions $\mathbf{p} = \gamma_u m\mathbf{u}$ and $E = \gamma_u mc^2$ and applying the relativistic velocity-addition rule of Lecture 16, one finds after some algebra that the momentum and energy measured in O' are

$$\boxed{p'_x = \gamma_v\left[p_x - \frac{v}{c}\left(\frac{E}{c}\right)\right], \qquad p'_y = p_y, \qquad p'_z = p_z, \qquad \frac{E'}{c} = \gamma_v\left[\frac{E}{c} - \frac{v}{c}\,p_x\right].} \tag{24}$$

Place these beside the position-time transformation of Lecture 13:

$$\begin{aligned} x' &= \gamma_v\Big[x - \frac{v}{c}(ct)\Big], & p'_x &= \gamma_v\left[p_x - \frac{v}{c}\left(\frac{E}{c}\right)\right], \\ ct' &= \gamma_v\Big[(ct) - \frac{v}{c}\,x\Big], & \frac{E'}{c} &= \gamma_v\left[\frac{E}{c} - \frac{v}{c}\,p_x\right]. \end{aligned} \tag{25}$$

The two sets are *identical* in structure, with $x \leftrightarrow p_x$ and $ct \leftrightarrow E/c$. The four quantities p_x, p_y, p_z, and E/c therefore transform exactly as x, y, z, and ct do.

Energy–momentum four-vector. The three momentum components together with the energy, (p_x, p_y, p_z, E/c), form a four-vector. It is also called the *four-momentum*.

Just as time (ct in proper units) is an inseparable part of the position-time four-vector, **energy is an inseparable part of the four-momentum**. Energy and momentum are bound together in the same way as time and space: an observer in a different frame mixes them into one another.

The invariant

Every four-vector carries an invariant — the time component squared minus the sum of the squares of the spatial components — that all observers agree on. For $A = (p_x, p_y, p_z, E/c)$ this is

$$A_t^2 - \mathbf{A}\cdot\mathbf{A} = \left(\frac{E}{c}\right)^2 - p^2. \tag{26}$$

Evaluate it in the most convenient frame: the one in which the particle is *at rest*. There $p = 0$ and $E = mc^2$, so the invariant is simply

$$\left(\frac{mc^2}{c}\right)^2 - 0 = m^2c^2. \tag{27}$$

Because the quantity is the same in every frame,

$$\boxed{\left(\frac{E}{c}\right)^2 - p^2 = m^2c^2 \qquad \Longleftrightarrow \qquad E^2 = p^2c^2 + m^2c^4.} \tag{28}$$

This is just the same the energy–momentum relation from Section 3: **the invariant of the energy–momentum four-vector is the mass**. Observers in different frames disagree about a particle's energy and about its momentum, but they all compute the same m. Mass is a property of the particle itself, not of the frame from which it is measured.

Why conservation laws work in every frame. The four-momentum also explains why the conservation statements for momentum (Lecture 17) and energy (Section 2) hold between frames. If the total four-momentum $\sum_i (p_{x,i},\, p_{y,i},\, p_{z,i},\, E_i/c)$ is unchanged by a collision in one frame, then because every component transforms linearly by the same Lorentz transformation, the total is unchanged in *every* frame. Conservation of momentum and conservation of energy are not two independent laws in relativity. They are the four components of a single law.

5 What Is *Not* Conserved: Mass Is Not Additive

The two sums above are the *only* ones guaranteed. In particular,

$$\sum_{i=1}^{N} K_i \qquad \text{and} \qquad \sum_{i=1}^{N} m_i \qquad \text{can both change.} \tag{29}$$

Kinetic energy can be converted into rest energy and back again. Since rest energy is mc^2, a reaction that converts kinetic energy into rest energy increases $\sum_i m_i$, and one that does the reverse decreases it.

Mass is not additive. The arithmetic sum $\sum_i m_i$ of the constituent masses is *not* conserved — but this does not mean mass is created from nothing. The *invariant mass of the system*, built from the totals via $M_{\text{sys}}^2 c^4 = \left(\sum_i E_i\right)^2 - \left(\sum_i \mathbf{p}_i\right)^2 c^2$, *is* conserved, because $\sum_i E_i$ and $\sum_i \mathbf{p}_i$ separately are. What fails in relativity is the Newtonian assumption that a system's mass is the sum of its parts' masses: a system's mass also includes the kinetic and binding energy of its parts.

In an inelastic collision kinetic energy disappears and the product is *heavier* than its parts, while bound nuclei are *lighter* than their separated constituents.